Dry Ice: Simply Sublime!



Background:

Dry Ice is frozen carbon dioxide, a gas which makes up about .04% of our atmosphere. As a gas, it is colorless and odorless, and is released when living organisms respire. It is also 1.5 times the density of air, (which is 78% nitrogen, 21% oxygen, and less than 1% argon.) Carbon dioxide (CO_2) gas turns to dry ice when the temperature drops below 78.5° C. It turns to vapor directly from the solid state, never passing through a liquid phase. This change, from a solid to a gas, is called "sublimation."

Dry ice is largely used for refrigeration. Look in your Yellow Pages to find a local vendor. A couple of blocks about 2 inches by 10 inches by 7 inches cost about $10 in my area. Place it in an ice chest and keep it there, taking out only what you need. It is a good idea to wear gloves when handling the dry ice.

Students are familiar with water ice. Dry ice gives you a chance to challenge their expectations, and force them to create some new models to account for their observations. Observation and experimentation of dry ice sheds light on two concepts: changes of state, and the behavior (or even existence!) of gases. Here are some steps you might follow to guide your students in uncovering the properties of dry ice in particular, and matter in general.

Step One: What is a Solid?

Experiment with water ice to establish melting point and boiling point. It is useful for students to graph the temperature as ice is heated, melted, then boiled. They will observe a graph which looks levell for a while, then reflects the temperature of the water rising, and then levels off again when it reaches the boiling point.

The water stays at 0° C as long as there is ice left to absorb the heat by melting. The liquid water warms to boiling, then again holds steady, so long as water remains in the liquid phase, absorbing heat by turning to vapor. Activities which do this are widely available, so I will not repeat them. It might be useful to relate the idea of solids melting to other materials they are familiar with. Water ice is one solid which melts. Others include metals, chocolate, wax, and plastic. Water melts at 0° C, mercury melts at 39° C, gold at 1064°C.

Step Two: Discovering Sublimation

Give the kids a chunk of dry ice, one to two inches square. Warn about possibledanger from contact. (Since dry ice is so cold, it will quickly freeze skin if it is in contact for more than a few seconds. The result is similar to a burn, though no heat is involved) Give them also a styrofoam cup, and a

piece of plastic wrap. Have them cover the cup, and observe. As the ice sublimes, it will swell the plastic, and may even pop it, if they can maintain a tight enough seal. They should record their observations.

Note: This will look more dramatic with water in the cup, but I have found that the kids often mistake the vapor for steam, and get the idea that the ice is actually "boiling" the water, a wrong idea, since the bubbles coming up are mainly CO2, not water vapor. If students blow into the cup, they will see even more white vapor. Why is that? Isn't it the same as hot steam? No, it isn't. The dry ice vaporizes into a clear, colorless gas. The white vapor we see is actually water in the air which is condensing. Our warm, moist breath cools quickly when in contact with the dry ice, and as it cools, the moisture condenses into tiny, fog-like droplets. It is like steam, in that it is water vapor, but it is not the product of heated water.

Water ice will sublime as well. Ice cubes will shrink when left in the freezer, frozen foods will become desiccated.

Step 3: What is this gas?

After they have experienced sublimation without water, and you have established that the dry ice is indeed turning into a gas, you could explore what the gas is. They may notice that the vapor coming off their cup goes down, unlike hot steam, which rises. This is due both to the temperature difference, and the inherent density of CO2 gas. If the kids place a lit candle on the table, they can then "pour" the CO2 gas from their styrofoam cup onto the flame. It should go out. This is evidence that you have a gas, heavier than air, and that it is a gas which does not provide oxygen for combustion. CO2 fire extinguishers are quite common.

You might point out that this gas is the same one which living things respire (breathe out). In high concentrations, it is poisonous to life, as well as to fire. It is also the gas released by Alka Seltzer and the gas in the bubbles of soda pop.

Note: The kids have to "pour" very cautiously, as a slight breeze will blow the CO2 away

Step 4: Experimenting with Dry Ice and Water.

Give them water and a thermometer. Tell them not to stick the thermometer right on the dry ice, and not to get it stuck in the water ice which forms.

You have established that the dry ice is shrinking, and turning into a gas. What happens when you add water? They will see a high rate of bubbling at first. They should note the rate of bubbling, because they should discover that it slows as the water cools. When the water cools enough, water ice will form a covering on the dry ice. This is interesting to observe. The ice will even encapsulate the chunk of dry ice, then pop, as further

sublimation builds up inside the capsule of ice. They might experiment by switching the water after it has gotten very cold.

After students have experimented in an open-ended way, ask them to measure temperature in a systematic way. They should begin with a new chunk of dry ice, and water at room temperature. They should measure the temperature of the water before they add the dry ice, then record the temperature every minute after the dry ice is added. They should create a graph similar but opposite to the one they created of the ice melting and boiling. These graphs illustrate that heat is being transferred through the process of the phase change. (So long as there is liquid water with the dry ice, it can transfer heat to the dry ice as it freezes. So long as there is liquid water being heated, it can transfer heat to the gas it forms as steam. Thus the temperatures remain at the freezing point or boiling point as long as liquid remains)

Remind them that the water is not "boiling," even though it is bubbling. Remind them also that the "steam" they see, is actually CO_2 gas made white by water vapor.

A question you might pose to the students is: Does the CO_2 sublime more quickly in water than air? Why?

[Answer: Water transfers heat much more quickly than does air at the same temperature. Drawings illustrating the kinetic model of gas and liquid would be useful to see why this is so.]

Bubble Chamber Fun

At the Exploratorium in San Francisco there is The Bubble Chamber. It is a large, four foot high plexiglass cylinder, about two feet in diameter. At the bottom of this cylinder there is a large chunk of dry ice. After it has been subliming there for a while, a layer of CO_2 gas builds up in the lower part of the cylinder. If you blow soap bubbles into the cylinder, they drop down to the level of the CO_2 gas, and float there, since the air within them is less dense than the CO_2 gas. Large plexiglass cylinders are hard to find, so just blow the bubbles into the ice chest where you have your dry ice. You will need to be careful to avoid breezes which would displace the gas, and you will need a fairly big chunk of dry ice to do this.

Get the students to develop hypotheses and conduct research to find out why the bubbles float. They should be able to find out in the library that CO_2 gas is denser than air. Their earlier experiment, where they "poured" the CO_2 gas onto the candle also supports the idea that CO_2 gas is denser than air.

Student Directions: You might wish to transfer this onto a worksheet with lines for student writing.

Dry Ice: Simply Sublime: Part One

Warning: Handle Dry Ice with caution. It can quickly freeze your skin! Anyone playing during the lab activity will not be allowed to participate.
I. Take a chunk of dry ice, no bigger than 4 cm across. Place it in a styrofoam cup and experiment with it. Record what you see.
2. Break the chunk of dry ice up into smaller pieces by poking it with a pen or pencil. Cover the opening of the cup with a piece of plastic wrap. Record what you observe.
3. How do you explain your observations so far? What is "Dry Ice"? What is it doing?
4. Light a candle, and carefully pour the "air" from your cup onto the flame. Do not let any solid dry ice come out. What happens?
5. How would you explain this?
6. Add water to your cup. What do you observe?
The following questions should get students to critically reflect on their observations.

Dry Ice Puzzlers

1. When you pour the gas from the dry ice onto the candle, why does it go down?
2. If the gas from dry ice is colorless and clear, why does it form clouds of white vapor, especially when you blow on it?
3. When you add dry ice to water, it seems to boil. Why is that?
4. The dry ice in water "pops" every once in a while. Explain what is happening.
5. Does the dry ice disappear more quickly in air or in water? Can you explain why?
6. While the dry ice was in the water, the temperature fell to 0° but no farther. Why do you think that is?
7. What temperature do you think the dry ice is?
8. In the Bubble Chamber, air bubbles floated. Can you explain why?
9. Is Dry Ice a good name for this material? Why or why not?